Bernardo Rivas

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Department of Mathematics and Statistics
University of Toledo
Toledo, OH 43606, USA

Employment

University of Toledo — Ohio, US
Postdoctoral Fellow, Department of Mathematics and Statistics — 2025 – Present
Supervisor: William Kalies
Supported by AFOSR MURI, in collaboration with the University of Michigan

Education

Rutgers University — New Jersey, US
PhD in Mathematics, Advisor: Konstantin Mischaikow — 2019 – 2025

Universidade de São Paulo — São Paulo, Brazil
MS and BS in Mathematics, MS Advisor: Marcio Gameiro — 2012 – 2019

Publications

Topological Dynamics via Learned Hybrid Systems — 2026
B. Rivas, K. Iwasaki, W. Kalies, A. Bloch, M. Ghaffari
Proceedings of the 8th Annual Learning for Dynamics and Control Conference (L4DC),
PMLR 331:1663–1674

Preprints

Computation of Attractor Lattices for a Class of Hybrid Systems — 2026
W. Kalies, **B. Rivas**, T. Wehbe
arXiv:2610.15123

Characterizing High-dimensional Dynamics by Combinatorial-Topological Methods on a Latent Space — 2026
P. Bailon, M. Gameiro, B. Gelb, W. Kalies, M. Kramar, K. Mischaikow, **B. Rivas**, E. Vieira
arXiv:2609.01509

Conley Index Theory for Hybrid Systems — 2026
B. Rivas, W. Kalies
arXiv:2606.18501

Boolean Models Coarsely Sample Continuous Dynamics of Regulatory Networks — 2026
B. Cummins, M. Gameiro, T. Gedeon, K. Mischaikow, **B. Rivas**
arXiv:2606.14925

Global Dynamics of Ordinary Differential Equations: Wall Labelings, Conley Complexes, and Ramp Systems — 2024
M. Gameiro, T. Gedeon, H. Kokubu, K. Mischaikow, H. Oka, **B. Rivas**, E. Vieira
arXiv:2412.11078

In Preparation

Learning Cycling Signatures of Hybrid Systems
D. Hien, K. Iwasaki, **B. Rivas**

Topological Validation of Parameter Inference in Gene Regulatory Networks
W. El-Khateeb, W. Kalies, **B. Rivas**

Towards an Algorithmic Approach to Control of Dynamic Phenotypes
M. Gameiro, W. Kalies, K. Mischaikow, **B. Rivas**

Talks

Characterizing global dynamics of hybrid systems — New Brunswick, NJ
Applied Math Seminar, Rutgers University — 2026

Homological Dynamics — New Brunswick, NJ
Conference on Algebra, K-theory and Applications — 2026

Computational Dynamics — Toledo, OH
UToledo Mathematics Colloquium — 2026

Computational Conley Theory — Vitória, Brazil
UFES – Universidade Federal do Espírito Santo — 2026

A computational topology approach to gene regulation — Austin, TX
Texas-Louisiana Section of SIAM 8th Annual Meeting — 2025

Global dynamics in systems biology — Denver, CO
SIAM Dynamical Systems — 2025

Computational dynamics of gene regulatory networks — Athens, OH
Dynamics Seminar — 2025

Network dynamics via combinatorics, topology and homology — Toledo, OH
AMS Seminar — 2025

Analytical bounds for combinatorial topological dynamics — Montreal, Canada
Workshop – Combinatorial topological framework for nonlinear dynamics — 2024

A computational framework for dynamics of genetic regulatory networks — Washington, DC, US
AMS Spring Eastern Sectional Meeting — 2024

Combinatorial topological method for gene regulatory networks — Leiden, Netherlands
Workshop – Computing combinatorial dynamics in high dim. biological networks — 2023

Rigorous numerics of invariant manifolds — São Carlos, Brazil
Summer Meeting in Differential Equations — 2019

O Jogo da Vida (The Game of Life) — São Carlos, Brazil
Simpósio de Matemática para a Graduação (SiM), ICMC-USP — 2018

O Jogo da Vida (The Game of Life) — São Carlos, Brazil
Seminário de Coisas Legais, ICMC-USP — 2017

Teaching

Course Development

DATA 5330 – Data Analytics and Visualization Online M.S. in Data Science and AI	University of Toledo	2026 – 2027

Instructor of Record

DATA 4980 – Advanced Topics in Data Science	University of Toledo	Spring 2026
640:311 – Introduction to Real Analysis I	Rutgers University	Summer 2024
640:300 – Introduction to Mathematical Reasoning	Rutgers University	Summer 2023
640:252 – Elementary Differential Equations (online)	Rutgers University	Summer 2022
Minicourse on Differential Equations	ICMC-USP	Winter 2019

Teaching Assistant (Rutgers University)

640:311 – Introduction to Real Analysis I	Spring 2024
640:311 – Introduction to Real Analysis I	Fall 2023
640:244 – Differential Equations for Engineering and Physics	Spring 2023
640:481 – Mathematical Theory of Statistics	Fall 2022
640:421 – Advanced Calculus for Engineering	Fall 2022
640:244 – Differential Equations for Engineering and Physics	Spring 2022
640:151 – Calculus I for the Mathematical and Physical Sciences	Fall 2021
640:135 – Calculus I for the Life and Social Sciences	Summer 2020

Mentoring

Informal research mentoring (not thesis advisor of record):

Wael El-Khateeb (now Assistant Professor, Emirates Aviation University) — University of Toledo
Topological validation of parameter inference in gene regulatory networks. — 2025 – 2026

Tony Wehbe — University of Toledo
Computational Conley theory for hybrid systems. — 2025 – Present

Software

Hybrid BoxMap: Conley-Morse graphs for hybrid systems — Python

MorseGraph-L4DC: Convex optimization and topological analysis for hybrid systems — Python

LatentDynamics: Topological analysis of learned dynamics on latent spaces — Python

DSGRN: Dynamic Signatures Generated by Regulatory Networks (core algorithms) — C++/Python

CMGDB: Conley-Morse Graph Database — C++/Python

Poster Presentations

L4DC: Learning for Dynamics and Control, Los Angeles, CA, USA — 2026
CBMS: Applied Mathematics and Machine Learning, Houston, TX, USA — 2025
TRIPODS/DATA-INSPIRE Graduate Student Workshop, DIMACS, Rutgers University — 2022
Workshop of Thesis and Dissertations, ICMC-USP, Brazil — 2018
Algebraic Topology in Data and Dynamics, Montana State University, USA — 2018

Service

Minisymposium organizer, SIAM Mathematics of Data Science — 2026
Referee, *Nonlinear Analysis: Hybrid Systems* — 2026
Mentor, Directed Reading Program, Rutgers University — 2024
Member, Graduate Student Liaison Committee, Rutgers University — 2022 – 2025
Head Proctor, Rutgers University — 2021 – 2023
Poster judge, Simpósio de Matemática para a Graduação (SiM), ICMC-USP — 2018
Undergraduate representative, Mathematics Department Committee, USP — 2015 – 2016

Conference & Workshop Participation

SIAM Annual Meeting, Cleveland, OH, USA — 2026
Modularity of Biological Systems Workshop, NITMB, Chicago, IL, USA — 2026
Id. of Gene Regulatory Network Function from Data, Providence, RI, USA — 2022
Simpósio de Matemática para a Graduação (SiM), ICMC-USP, Brazil — 2020
Connections in Infinite Dimensional Dynamics, BIRS — 2020
International Congress in Mathematics, Rio de Janeiro, Brazil — 2018
Summer Meeting in Differential Equations, ICMC-USP, Brazil — 2017
GeoGebra Global Gathering, Linz, Austria — 2015

Fellowships

Research Internship Abroad, FAPESP – Rutgers University — 2018
Master's Fellowship, FAPESP – Universidade de São Paulo — 2017 – 2019
Science Without Borders, CNPq – Queen Mary University of London — 2014 – 2015
Scientific Initiation Fellowship, FAPESP – Universidade de São Paulo — 2013 – 2016